

BT 3041 Analysis and Interpretation of Biological Data

Assignment 1

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All the relevant code for the problems solved as part of this assignment can be found here:
<https://colab.research.google.com/drive/1li-wSPYHa-1olrmeMGjVYjWbAFFsVUxP?usp=sharing>

The Google Drive folder containing all relevant data, code and the report can be found here:
https://drive.google.com/drive/folders/1B7V8eDYZudrjEXTBxzdBI7LWgSN5VZ2u?usp=drive_link

1. Problem Statement 1:

a. Data Extraction:

- We initially access the spreadsheet containing the data from Google Drive, and extract the sheet titled 'Blood Glucose' for the purposes of our analysis. The 0th row, comprising the titles of different columns, is used to assign titles, and is dropped from further analyses; subsequently, the row indices are accordingly changed.
- We then separate the data corresponding to male and female mice, across all weeks when measurements were taken, as required by the problem statement. We subsequently obtain only the relevant data (i.e. data from week 3, 5, 7, 9, 11, 12, 13, 14, 15, 16, and the column that represents the sex to which the mouse specimen belongs) in a separate pandas dataframe. 50 male mice and 60 female mice specimen were used to obtain these data.
- These data are used to plot histograms of blood glucose distributions of male and female mice (plots corresponding to the same week overlaid in the same graph) from week 3 to week 16, using the seaborn function `seaborn.histplot()`. A customized palette was used to visualize the histograms better.

b. Interpretation of the plots (see Figure 1):

- We now seek to determine which of these plots follow a normal distribution, and which do not, based on the symmetry of the histograms, with the help of overlaid Kernel Density Estimate (KDE) plots.
- Assumptions:
 - i. While studying the histograms, we do not overlook outlier values (which may be artefacts of small sampling), such as in the distributions for female mice on weeks 15 and 16.
 - ii. In some cases, due to the absence of data on either side of the absolute maximum (such as the distribution of female mice on week 11), we use both the histogram and the KDE plot to determine whether the plot follows symmetry, and, thus, claim that it approximately follows a normal distribution.
- From the histogram and the KDE plots, we conclude that the blood glucose distribution for female mice follows a normal distribution on week 3. The blood glucose distribution for male mice also appears to follow a normal distribution. However, it is not clear whether the distribution for male mice has a broad maximum around blood glucose values 120 to 160 (indicating high variance and standard deviation), or whether a sharp maximum somewhere between these two values has been obscured by the relatively small size. Furthermore, this observation could also be explained by the presence of a left-skewed distribution with a long left tail, even as the KDE plot indicates a more symmetric plot. Thus,

based on the histogram alone, we conclude that the distribution for male mice does not follow a normal distribution.

- For the data on week 5, the distribution for male mice appears to have a long right tail, indicating a right-skewed distribution (again, with a broad maximum around blood glucose values 120 to 160); however, the KDE plot seems to suggest a more symmetric, normal distribution. The distribution for female mice has a wide left tail, indicating a left-skewed distribution. A relatively small peak (local maximum) around blood glucose value 90, along with a broad left tail (indicating a left-skewed distribution) also suggests that a normal distribution is not followed by the distribution for female mice.
- On week 7, the distribution for male mice follows a normal distribution, this time with relatively smaller variance and standard deviation as compared to the distributions on weeks 3 and 5, indicated by a relatively sharper maximum. While the distribution for female mice is symmetric, a relatively small peak (compared to the maximum), near the blood glucose value 80 suggests that a normal distribution is not followed.
- On week 9, the distribution for female mice is bimodal in nature, given the two local maxima observed near blood glucose values 110 and 140, indicating a normal distribution is not followed. Furthermore, the distribution for male mice lacks symmetry about the maximum, and has a relatively small peak (compared to the maximum), near the blood glucose value 150, indicating that it does not follow a normal distribution.
- Based on the symmetry of the histogram and the KDE plot, the distribution for female mice follows a right-skewed distribution (indicated by the long right tail) on week 11, even though we do not observe the left tail due to a lack of blood glucose values less than 80, and the KDE plot seems to suggest a more symmetric distribution. This observation could also be due to the low frequency of outliers with relatively high magnitude. The distribution for male mice lacks symmetry about the maximum, indicated by a relatively small peak (compared to the maximum), near the blood glucose value 150.
- On week 12, the distributions for male and female mice have a relatively small overlap (compared to other distributions in this problem). Based on the symmetry of the histogram and the KDE plot, the distribution for male mice follows a normal distribution, even though we do not observe the right tail due to a lack of blood glucose values greater than 280. The distribution for female mice is not symmetric, and, thus, cannot be considered as a normal distribution; the presence of a long right tail indicates a right-skewed distribution.
- On week 13, based on the histograms, neither of the two distributions follow the symmetry that is the property of a normal distribution. Furthermore, based on the KDE plot, the distribution for female mice has a long right tail (indicating a left-skewed graph), the distribution for male mice has a long left tail (indicating a right-skewed graph).
- On week 14, the distributions for male and female mice, again, have a relatively small overlap (compared to other distributions in this problem). Here, both the distributions appear to follow normal distributions, even as the left tail from the distribution of female mice and the right tail for that of male mice are not observed.
- Based on the histogram, on week 15, the distribution for female mice has a long right tail, indicating a right-skewed distribution. The lack of symmetry, and the presence of a local maximum around blood glucose value 150, apart from the absolute maximum of the plot, indicates that the distribution for male mice does not follow a normal distribution.
- On week 16, the distribution for female mice has a long right tail, indicating a right-skewed distribution. On the other hand, the distribution for male mice lacks symmetry, and has a relatively short peak (apart from the absolute maximum)

close to the blood glucose value 250, indicating the distribution for male mice does not follow a normal distribution. The long right tail is indicative of a right-skewed distribution.

- We have thus ascertained, based on the assumptions stated above, whether the blood glucose distributions for male and female mice sampled on specific weeks follow a normal distribution, or otherwise.

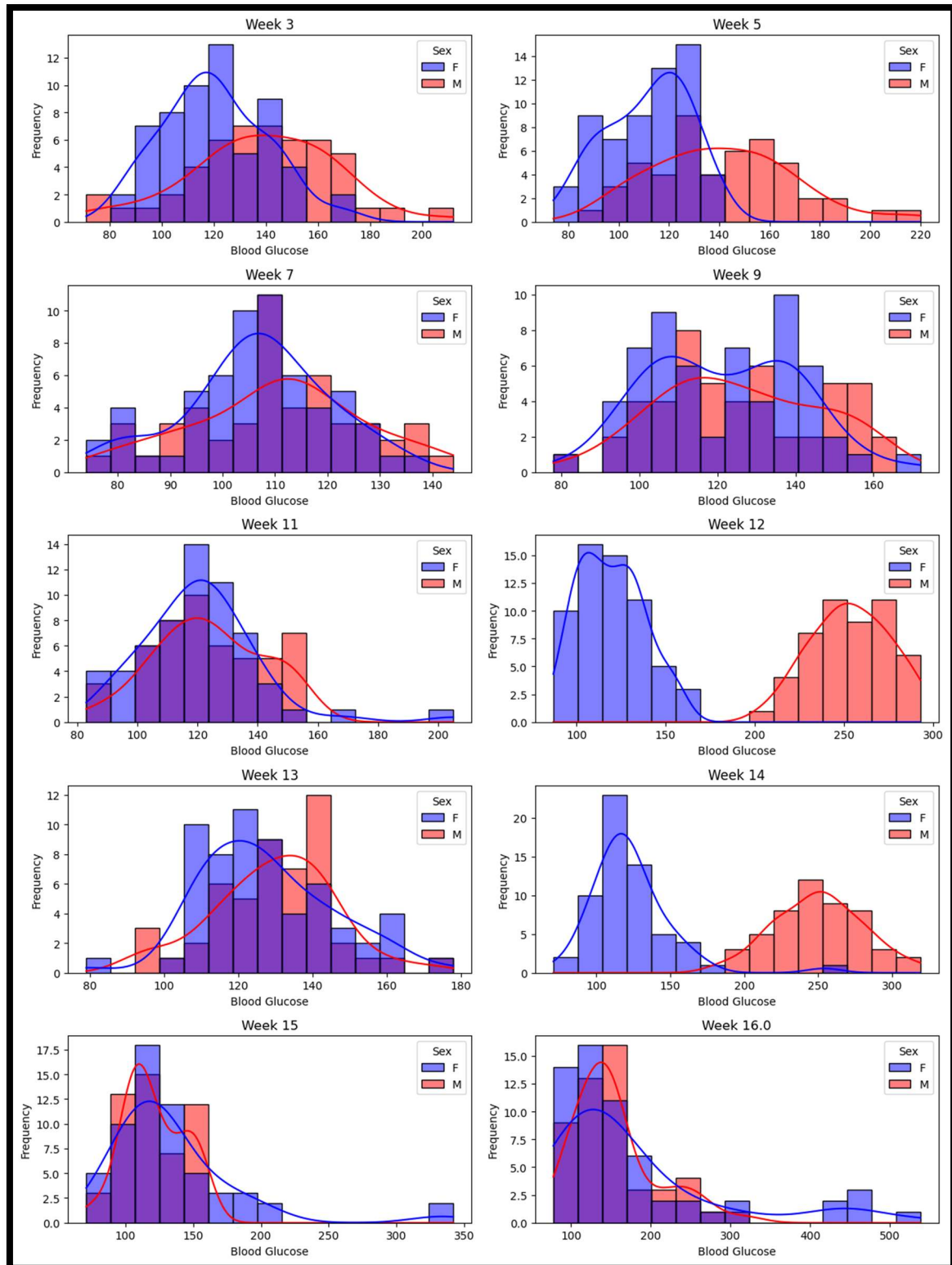


Figure 1: Histograms to visualise the blood glucose distributions of male and female mice across different weeks.

2. Problem Statement 2:

a. Data Extraction:

- The data for blood glucose counts of male and female mice across specific weeks, as extracted above, was used for plotting the box plots.
- Data required for specific early age (weeks 3, 5 and 7) and late age (weeks 14, 15 and 16) were obtained for each sex, and the box plots were plotted using the seaborn function `seaborn.boxplot()`.

b. Interpretation of the plots:

a. Early and Later Age Male mice (see Figure 2):

- The boxplot for male mice on week 3 indicates the distribution is left-skewed, owing to the left tail evident in the relatively wide 1st Interquartile Range (IQR). This is in agreement over the discussion provided in the solution to problem statement 1, where we could not confidently conclude whether the distribution was normal in nature. The source of this confusion could be traced to the presence of outliers to the left and right of the 1st and the 4th IQRs, respectively, (as seen in the boxplot) and the appearance of a broad maximum in the histogram, which may provide confounding results if referred only to the KDE plot.
- The boxplot for male mice on week 5 indicates the distribution is right-skewed, owing to the right tail evident in the relatively wide 4th Interquartile Range (IQR), and a outlier to the right of the 4th IQR. This is in agreement over the conclusions made in the solution to problem statement 1.
- We thus find that the histogram is more informative about the nature of the distribution than the KDE plot.
- The boxplot for male mice on week 7 indicates the distribution is slightly left-skewed, owing to a wider 2nd IQR, as compared to the 3rd IQR. However, the relative difference between the two seems to not have been captured in the histogram, thus leading to our earlier conclusion that this distribution is normal in nature. Sampling more specimen mice could indicate whether the relatively small difference between the 2nd and the 3rd IQR is an artefact that has arisen due to small sample size, or is due to the inherent nature of the distribution.
- In the early age mice, the medians across the three weeks follow the order: Week 3 $\approx$ Week 5 > Week 7.
- From the boxplot for male mice on week 14, the distribution of blood glucose appears symmetric, a conclusion which is in agreement with the one made in the solution to problem statement 1.
- From the boxplots for male mice on weeks 15 and 16, the two distributions are right-skewed, with distinctive right tails. In the case of week 16, there are 4 outliers, that lie to the right of the 4th IQR, indicating a highly skewed distribution. This conclusion agrees with the one made above in the solution to problem statement 1.
- In the later age mice, the medians across the three weeks vary substantially, with the more accurate order of means being Week 14 > Week 16 > Week 15.
- The median for week 14 is highest among all distributions of male mice considered here, while the other medians lie close to one another.

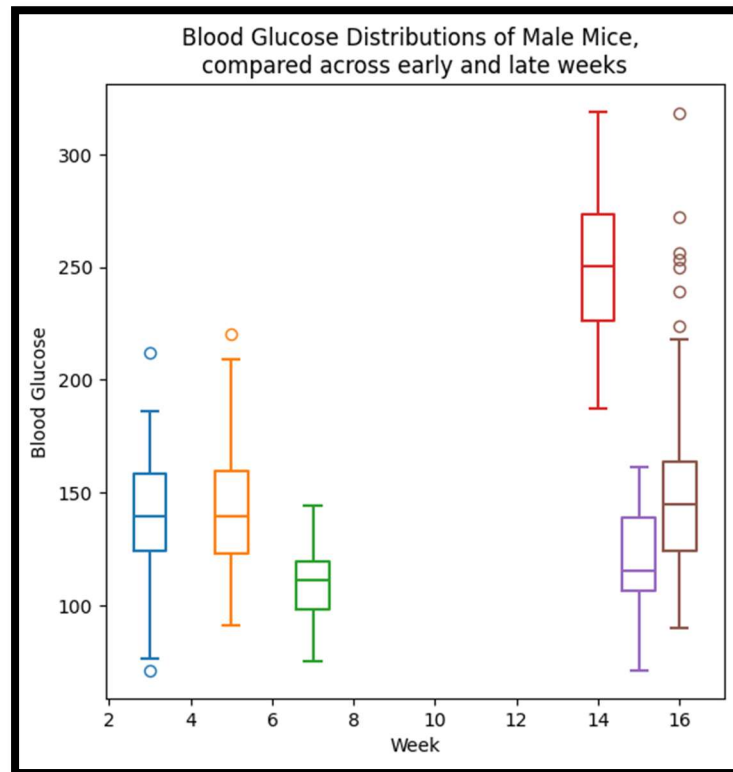


Figure 2: Boxplots to visualise the blood glucose distributions of early and late stage male mice

b. Early and Later Stage Female mice (see Figure 3):

- Among the early stage female mice, the blood glucose distribution on week 3 follows a normal distribution, based on the almost equal ranges occupied by all the 4 IQRs.
- The presence of a relatively wider 2nd IQR, as compared to the 3rd IQR, in the case of week 5, indicates the presence of a left-tail, and, hence, a left-tailed distribution.
- These conclusions agree with the ones made in the solution to problem statement 1.
- However, in the case of week 7, even as the distribution is normal in nature, due to the equal ranges occupied by all the 4 IQRs, the presence of an outlier on the left seems to have confounded our results in the solution to problem statement 1, where we concluded the distribution to not be normal in nature, due to the presence of a relatively short peak near the blood glucose value 80. It is this observation that shows up as an outlier close to the left end of the 1st IQR in the boxplot.
- In the early stage female mice, the medians across the three weeks follow the order, even though the values are close to one another:
Week 3 > Week 5 > Week 7
- Among the late stage female mice, all distributions are right-skewed in nature, due to the wide ranges occupied by the 3rd and 4th IQRs, in comparison to the 1st and 2nd IQRs.
- These results are in agreement, for the most part, with the conclusions made in the solution to problem statement 1, where we concluded that the distribution for female mice on week 14 follows a normal distribution. This inaccurate result may have arisen due to the number of 2 outliers present on the right of the 4th IQR,

and 1 outlier present to the left of the 1st IQR, as observed in the box plot. One of the outliers on the right is further away from the 4th IQR than the other one.

- In the late stage female mice, the medians across the three weeks follow the order: Week 16 > Week 14 > Week 15

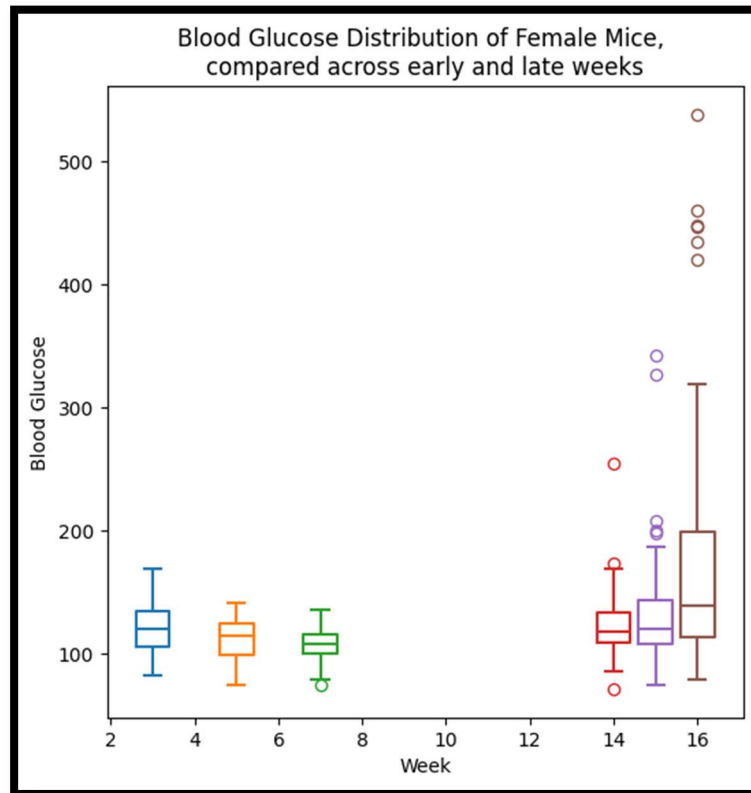


Figure 3: Boxplots to visualise the blood glucose distributions of early and late stage female mice

3. Problem Statement 3:

a. Data Extraction:

- The Z-scores across mice within each individual week can be obtained by calculating column-wise Z-scores within the spreadsheet titled 'Blood Glucose' in the dataset provided.
- The Z-scores across weeks within each individual mouse can be obtained by calculating row-wise Z-scores within the spreadsheet titled 'Blood Glucose' in the dataset provided.
- The first 60 rows represent female mice and the later 50 rows male mice specimen used to obtain the data. The columns refer to weeks numbered 3, 5, 7, 9, 11, 12, 13, 14, 15 and 16.

b. Interpretation of the plots:

a. Column-wise Z-scores (Figure 4):

- In the early weeks (week 3 to week 7), we see 'red' or 'peach' coloured elements in the bottom half of the Z-score heatmap, indicating that male mice have higher blood glucose more than female mice in the early weeks of development.
- These results agree with the observations from the histograms – the distributions for male mice shift from higher values to lower values, and align more closely with the distributions for female mice, from week 3 to week 7.
- In the later weeks (barring weeks 12 and 14), we see 'red' or 'peach' coloured elements in the top half of the Z-score heatmap, indicating that female mice have blood glucose values higher more than male mice in the later weeks of development.
- This transition is captured in the column for week 9, where we see the emergence of 'peach' coloured elements for the first time in the top half of the Z-score heatmap.
- In weeks 12 and 14, more 'peach' coloured elements are seen in the bottom half of the Z-score heatmap, which agrees with the results from the corresponding histograms, where we observe the male distributions shifted more towards higher values as compared to the female ones.

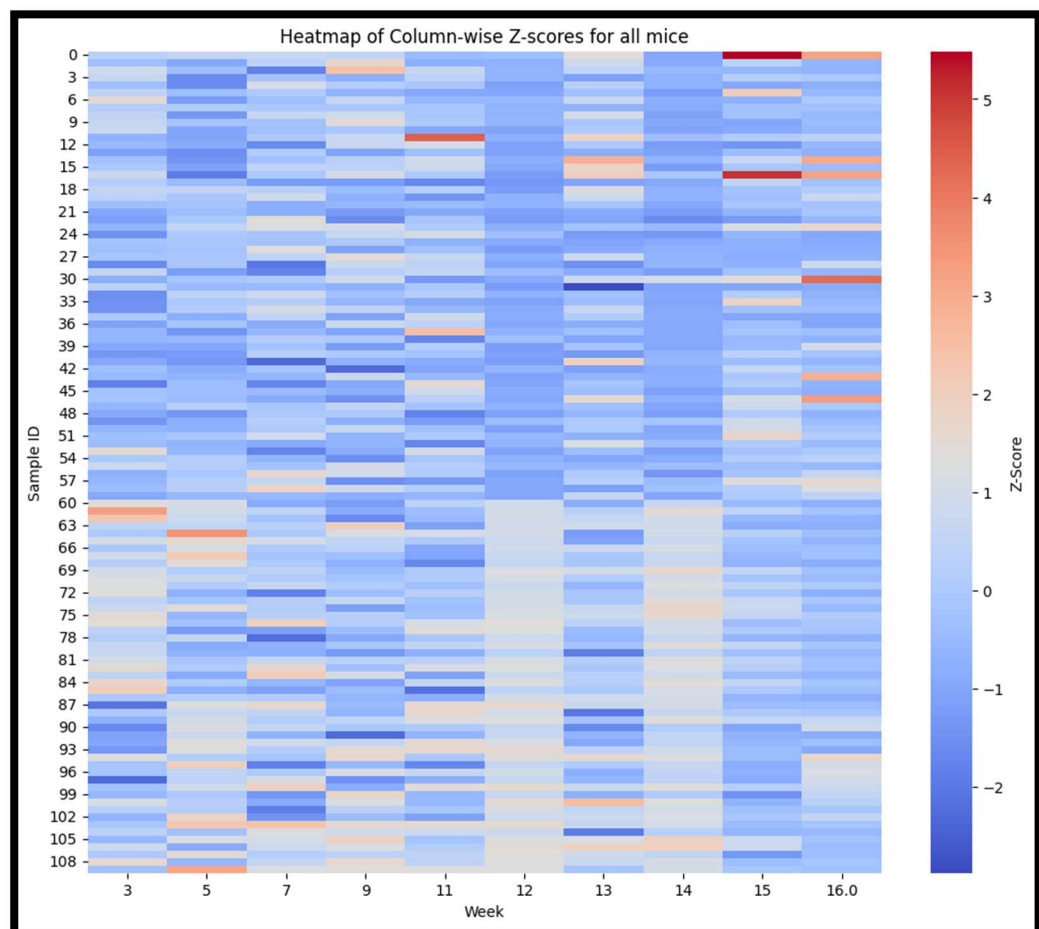


Figure 4: Heatmap depicting column-wise Z-scores for blood glucose values i.e. across mice within each individual week.

b. Row-wise Z-scores (see Figure 5):

- The row-wise Z-scores for female mice show the presence of 'dark red' coloured elements towards the right, or later weeks, indicating that older female mice tend to have higher blood glucose in the later weeks.
- The widespread presence of 'peach' and 'blue' coloured patches across rows in the weeks 7 and 14 indicates that the female mice seem to have reduced blood glucose during certain periods of the experiment. Thus, we need to investigate the environmental conditions to which the female mice were subjected, along with any changes in their diet, to ascertain the reason behind this apparent loss in weight at certain stage, across almost all female mice.
- A recent study in humans reported that blood glucose levels increase and decrease across certain phases of the menstrual cycle among females [1]; thus, the increase and decrease in blood glucose levels for female mice across weeks could also be associated with certain phases of their menstrual cycle.
- In the case of male mice, the 'red' coloured patches corresponding weeks 12 and 14 represent the weeks in which the mice, in general, reached their maximum blood glucose values, except certain mice, which saw an increase in their blood glucose values again from week 15 to week 16, indicated by 'red' coloured elements for week 16 in some mice specimen.
- These mice represent the outliers seen in the boxplot for male mice on week 16.
- This agrees with the histograms for week 12 and 14, where the distribution for male mice is seen to have shifted to higher blood glucose values, as compared to their female counterparts.
- The broad 'blue' coloured elements from week 3 to week 14 indicate that male mice do not gain blood glucose as regularly as female mice. Certain 'peach' coloured elements in some rows represent some mice that regularly gained blood glucose from week 3 onwards.
- Thus, both male and female mice see an increase in their blood glucose levels as they age from week 3 to week 16; however, we cannot determine the exact reasons behind certain periods where mice of both sexes saw reduced blood glucose levels, indicating the reasons may have more to do with factors like diet, environment, etc. than with genetic factors that regulate development.

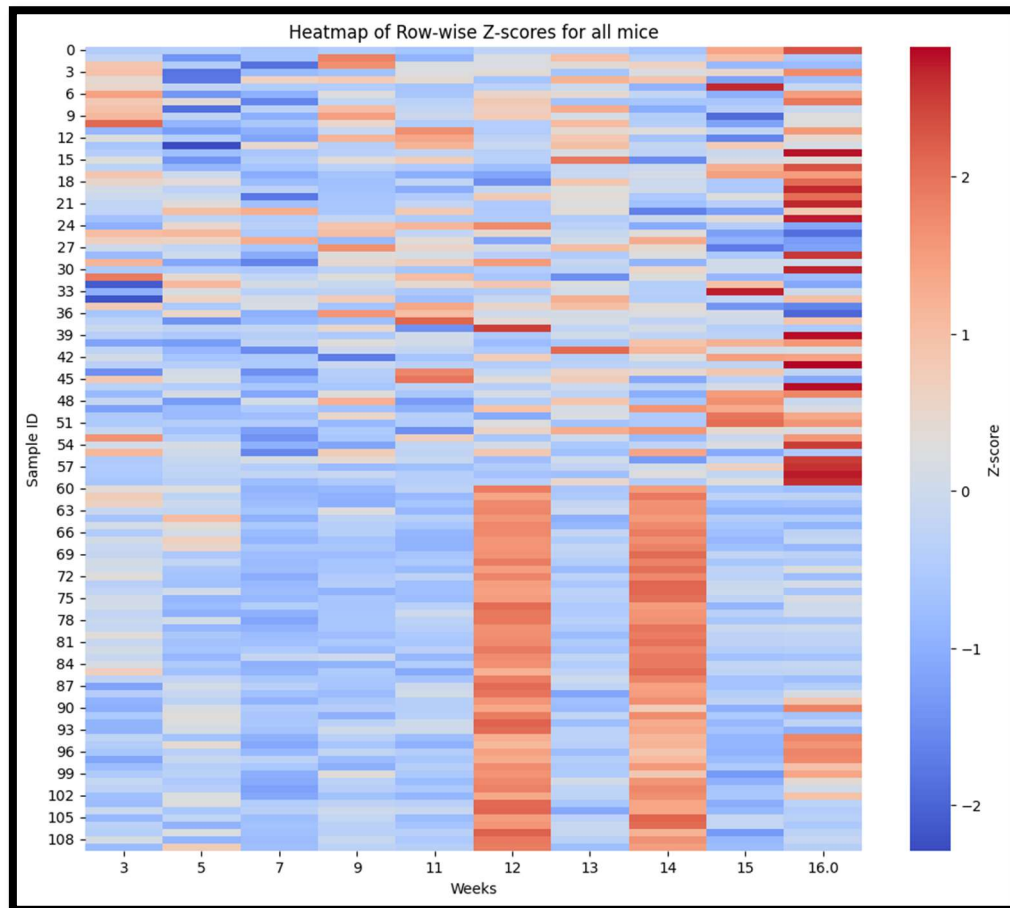


Figure 5: Heatmap depicting row-wise Z-scores i.e. across weeks within each individual mouse.

4. Problem Statement 4:

a. Data Extraction:

- The 'Clinical Chemistry' spreadsheet was extracted from the given dataset. Furthermore, the data for all mice from week 30 for the relevant metabolites/proteins were obtained from this spreadsheet.
- These data were used to plot the scatter plots for each of the pairs of metabolites/proteins mentioned in the problem statement, using the seaborn function `seaborn.scatterplot()`.
- The covariance matrix for each pair of metabolites/proteins was obtained using the pandas function `pandas.DataFrame.cov()`. An appropriate element of the covariance matrix (either [1,0] or [0,1], here) was used to report the required covariance.

b. Interpretation of the plots (see Figure 6):

- None of the scatter plots under consideration indicate the presence of a linear or polynomial function that can be used to explain how one of the variables varies with the pair.
- Certain pairs of metabolites/proteins form clusters. These include GLU and HDLX, GLU and ALT, ALB and HDLX, and GLU and CK. The data points for the TP versus BUN plot are relatively more widely scattered, and do not appear to form a cluster.

- The covariances for each of the pairs of metabolites/proteins are given as follows in **Table 1**:

Pairs of metabolites/proteins	Covariance in appropriate units
(a) Blood glucose level (GLU) and High Density Lipoprotein (HDLX)	490.5998461538462 (mg/dL) ²
(b) Blood glucose level (GLU) and Alanine Transaminase (ALT)	73.25230769230771 (mg/dL)(IU/L)
(c) Serum Albumin level and High Density Lipoprotein (HDL)	1.8025076923076926 (g/dL)(mg/dL)
(d) Blood glucose level (GLU) against Creatine Kinase (CK)	-3739.738461538461 (mg/dL)(IU/L)
(e) Total Protein (TP) against Blood Urea Nitrogen (BUN)	0.0981538461538462 (g/dL)(mg/L)

Table 1: Pairwise covariance values for the required pairs of metabolites/proteins

- In order to determine whether the variables are strongly or weakly correlated with each other, the Pearson Correlation coefficient for each of the pairs of metabolites/proteins can be determined.
- From the above table, we interpret that while most pairs of metabolites/proteins correlate positively with each other, GLU and CK correlate negatively with each other. GLU and CK also have the highest magnitude of covariance among all the pairs considered here, while TP and BUN have the smallest magnitude of covariance.
- We should be mindful of units while making this comparison, since while most units here denote concentration, IU/L denotes enzyme activity, which cannot be directly compared across different experiments that may use different assays or conditions to determine enzyme activity.
- The various scatter plots, as required, are given as follows:

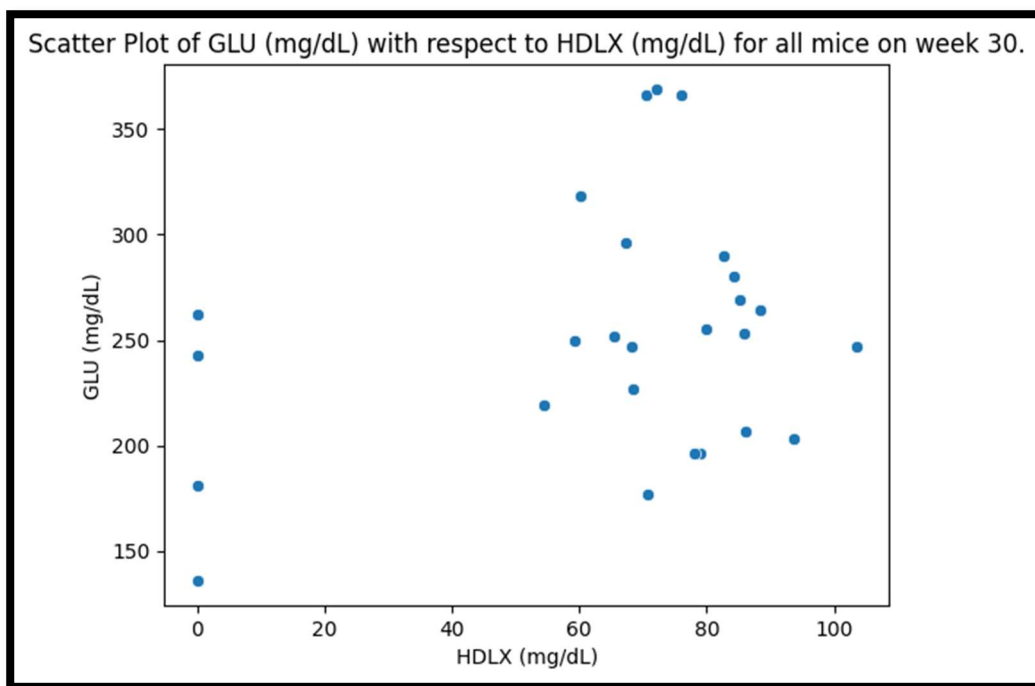


Figure 6 (a): Scatter plot of GLU (mg/dL) against HDLX (mg/dL) for all mice on week 30.

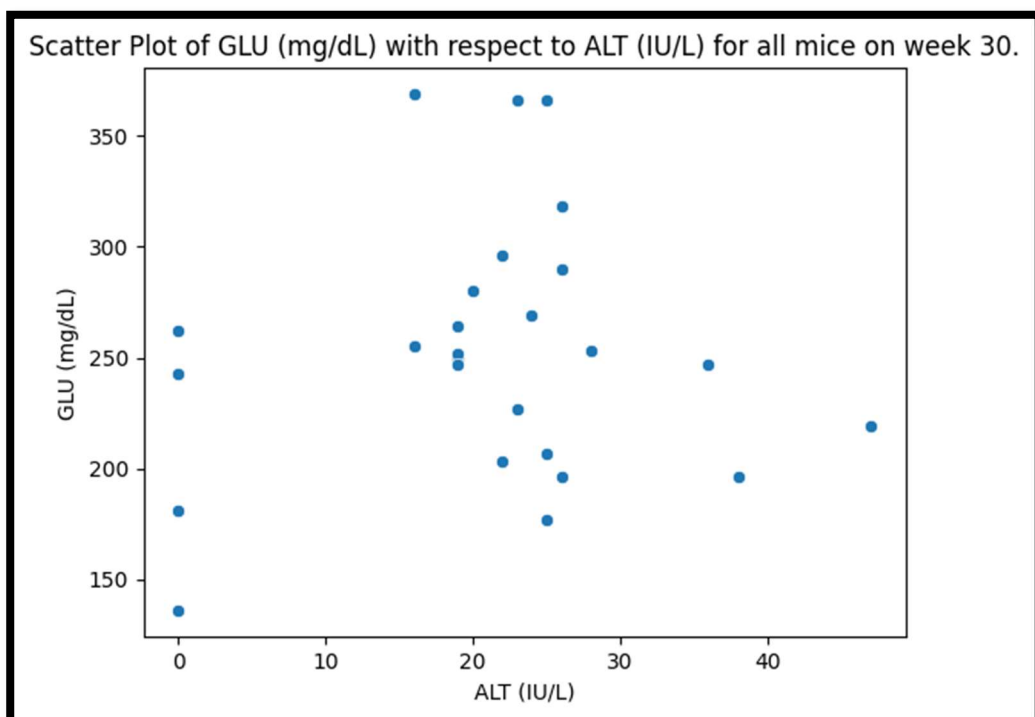


Figure 6 (b): Scatter plot of GLU (mg/dL) against ALT (IU/L) for all mice on week 30.

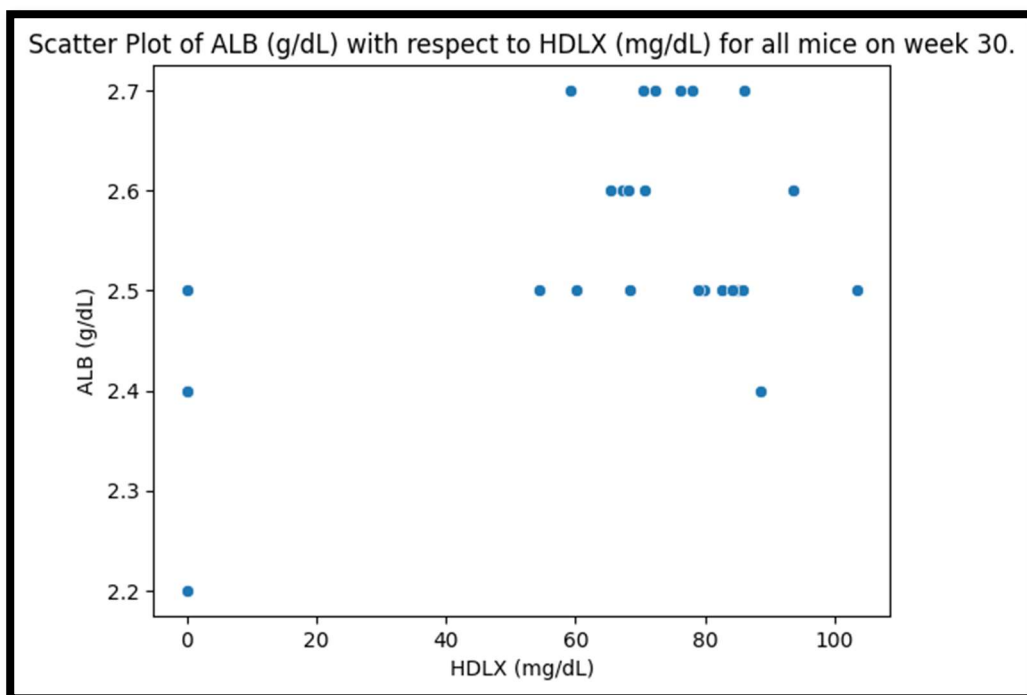


Figure 6 (c): Scatter plot of ALB (g/dL) against HDLX (mg/dL) for all mice on week 30.

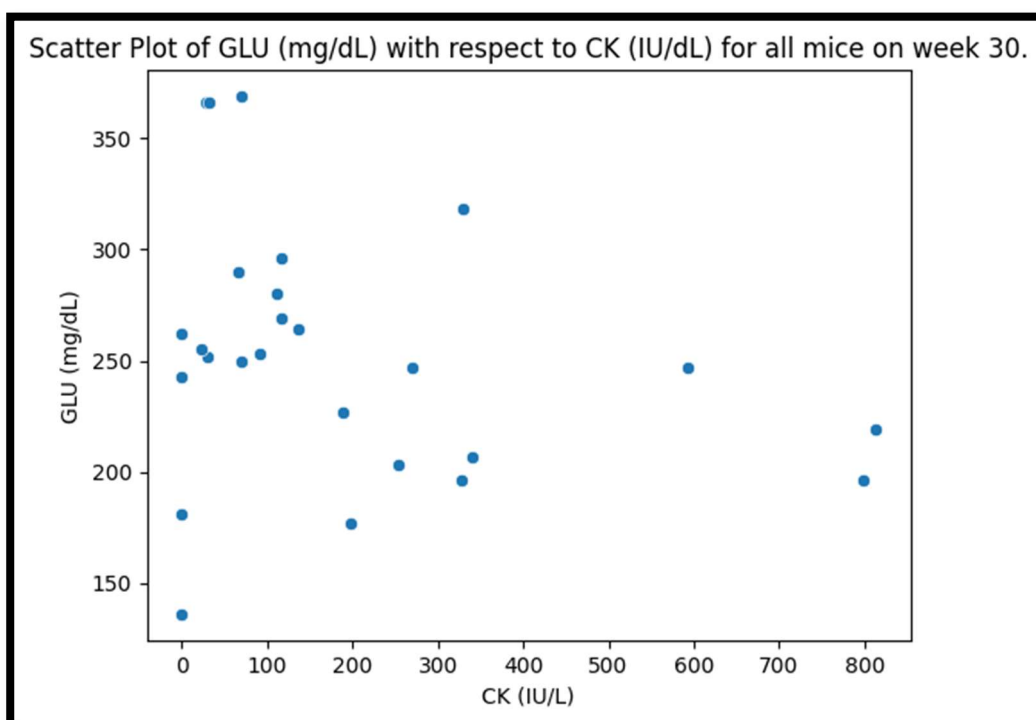


Figure 6 (d): Scatter plot of GLU (mg/dL) against CK (IU/L) for all mice on week 30.

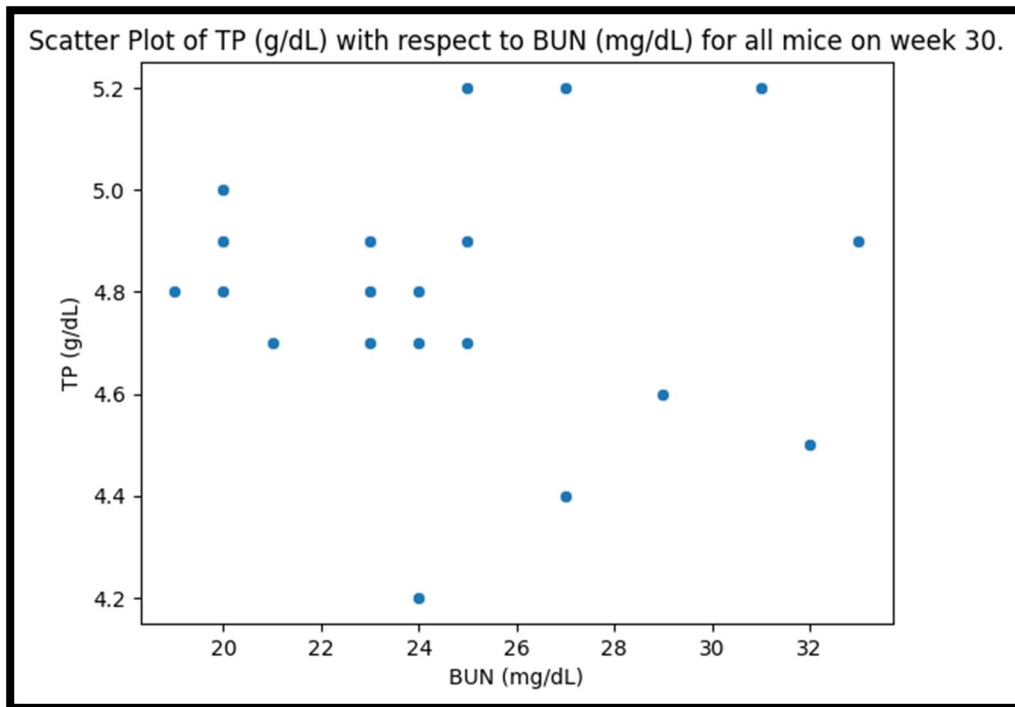


Figure 6 (e): Scatter plot of TP (g/dL) against BUN (mg/dL) for all mice on week 30.

5. Problem Statement 6:

a. Data Extraction:

- We extract the data for Fat Tissue (g) from the DXA spreadsheet, and further obtain the data for weeks 8 and 30 for male mice.
- These data are used for further analysis.

b. Statistical Test and Interpretation:

- Since we have been asked to evaluate whether the mean of a distribution changed significantly from one timepoint to the other, we must resort to a parametric t-test; we cannot compare parameters such as mean or variance using a non-parametric test.
- We first perform an F-test to determine whether the population variances of the two datasets significantly differ, and accordingly determine whether to use a separate-variance t-test or a pooled-variance t-test.
- The Null and Alternative Hypothesis for the F-test were as follows:

H₀: There is no significant difference in variance between the Fat Tissue (g) of male mice in Week 8 and Week 30.

H_A: There is a significant difference in variance between the Fat Tissue (g) of male mice in Week 8 and Week 30.

- The results of the F-test were as follows:

The results of the F-test are given as follows:

Variance of the Fat Tissue (g) distribution on week 8 = 0.23732689531056
Variance of the Fat Tissue (g) distribution on week 30 = 0.36
Calculated F value = 1.5168950806393562
Critical F value = 3.178893104458269

We accept (or, fail to reject) the null hypothesis with a p value = 0.2723249984491659, and a threshold p value = 0.05.

Figure 7: Results of the F-test

- Therefore, we conclude that the population variances of Fat Tissue (g) across Week 8 and Week 30 were not statistically significantly different since we fail to reject the null hypothesis.
- We now employ the Pooled-Variance t-test to determine whether their means are statistically significantly different (while we know the population variances do not vary significantly, we only know the sample variances, and thus cannot use the z-test).
- The Null and Alternative Hypotheses for the Pooled-Variance t-test were as follows:

H₀: The mean value of Fat Tissue (g) of male mice did not significantly increase from Week 8 to Week 30.

H_A: The mean value of Fat Tissue (g) of male mice significantly increased from Week 8 to Week 30.

- Based on the alternative hypothesis, we perform a one-tailed version of the t-test. The results of the Pooled-Variance t-test were as follows:

The results of the pooled-variance t test are as follows:

Mean of the Fat Tissue (g) distribution on week 8 = 6.5
Mean of the Fat Tissue (g) distribution on week 30 = 4.3535852
Pooled Variance = 0.5465010957493864

Calculated t value = 8.782286875619688
Critical t value = 1.7340636066175354

Therefore, we reject the null hypothesis with a p value = 3.1659582844234535e-08, and a threshold p value = 0.05.

Figure 8: Results of the Pooled-Variance t-test

- Therefore, since we reject the null hypothesis, we conclude that there was a statistically significant increase in the mean of Fat Tissue (g) from Week 8 to Week 30 in male mice.

6. Problem Statement 7:

a. Data Extraction:

- The 'Hematology' spreadsheet was extracted from the given dataset, and the WBC count data for all mice on week 30 (in $10^3 / \mu\text{L}$) were obtained from therein.

- The data were separated into those obtained from male and female mice, respectively, and then the following analysis were performed.

b. Interpretation:

- We first plot a histogram showing the distributions of WBC counts of male and female mice on week 30 to determine if they follow a normal distribution; a parametric or non-parametric statistical test can then be accordingly performed.

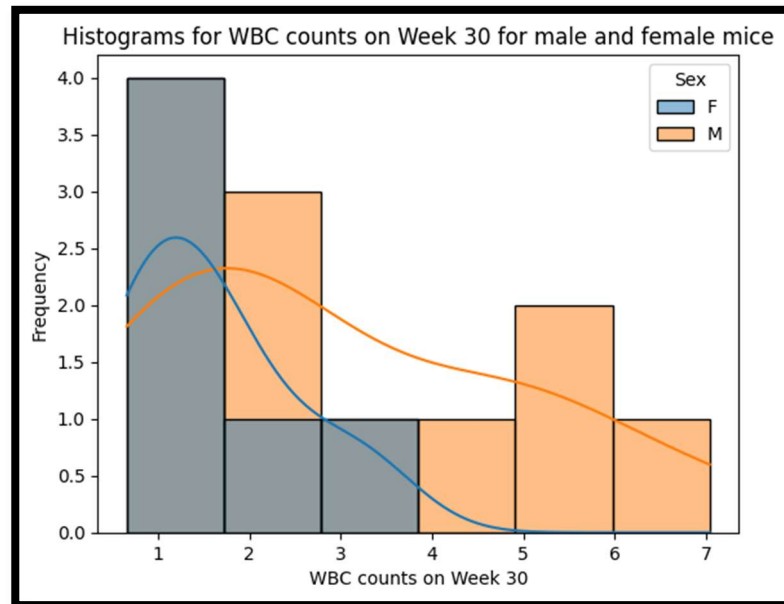


Figure 9: Histogram to visualise the distributions of WBC counts of male and female mice on week 30

- Based on **Figure 9**, it is clear that the distributions are not normal in nature. The distribution for male mice shows a long right tail (indicative of right-skewed data), while that for female mice shows a relatively shorter peak (from 5 to 6 x 10³ / μ L) in addition to the maximum, indicating the lack of symmetry in the distribution. However, it is not evident whether this result is due to a relatively small sample size (6 male and 12 female mouse specimen) or due to the inherent nature of the distribution.
- Therefore, we perform the Mann-Whitney U-test, which is the non-parametric equivalent of the t-test to determine whether there was a significant difference in WBC counts between male and female mice on week 30. We use the scipy function `mannwhitneyu()` from the stats package for this purpose.
- The Null and Alternative Hypotheses for the Mann-Whitney U-test are as follows:

H₀: There is no significant difference between the WBC counts of male and female mice on week 30.

H_A: There is a significant difference between the WBC counts of male and female mice on week 30.

- The results of the statistical test are as follows:

The results of the Mann-Whitney U-test are as follows:

The U Statistic = 50.5 and the p value = 0.18955480737931263

Therefore, we accept (or, fail to reject) the null hypothesis with a p value = 0.18955480737931263, and a threshold p value = 0.05.

Figure 10: Results of the Mann-Whitney U-test

- Thus, since we fail to reject the null hypothesis, we conclude that there was no statistically significant difference between the WBC counts of male and female mice on week 30.

7. Reference:

1. Lin, G., Siddiqui, R., Lin, Z. *et al.* Blood glucose variance measured by continuous glucose monitors across the menstrual cycle. *npj Digit. Med.* **6**, 140 (2023). <https://doi.org/10.1038/s41746-023-00884-x>